GUVI | HCL

Skill Up. Level Up

Train. Tune. Deploy.

Artificial Intelligence and Machine Learning Programme

6 Months

Live & Recorded Sessions Online

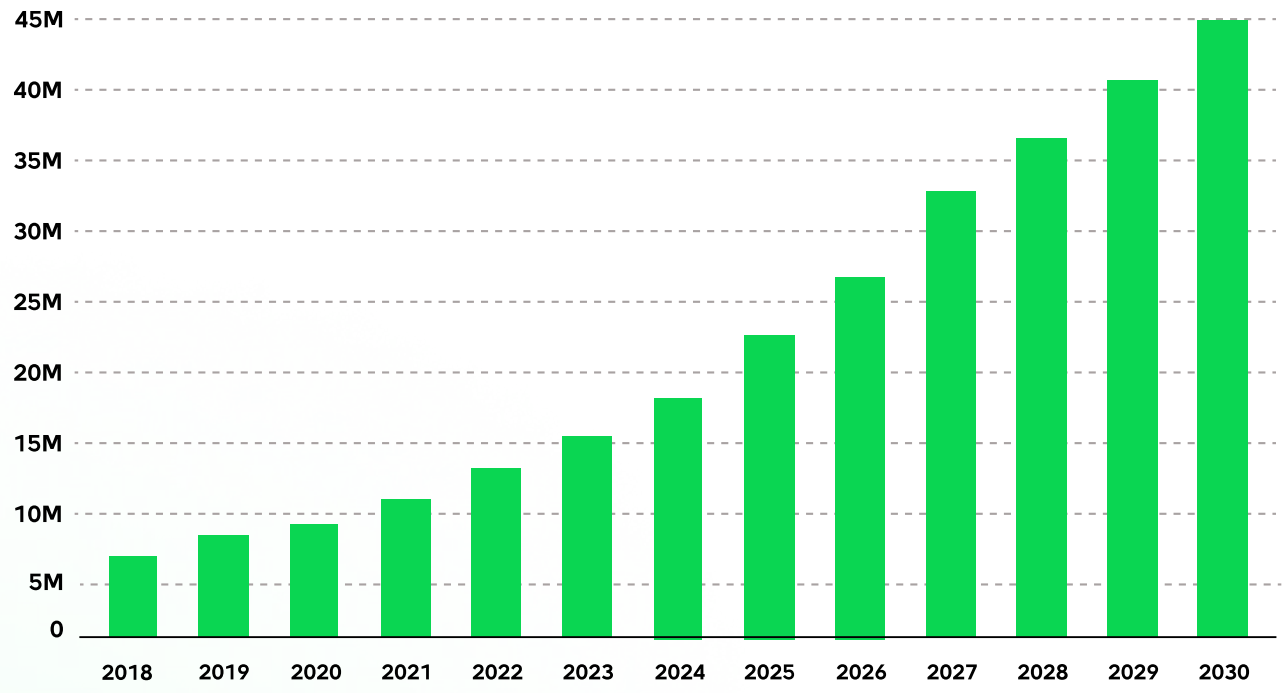
Weekend/ Weekday Classes Available

Available in English & தமிழ்

Become an AI/ML Engineer for the AI-Driven Era

Artificial Intelligence (AI) and Machine Learning (ML) are rapidly transforming industries and driving large-scale digital adoption. According to Grand View Research, the **global AI & ML market is projected to grow from USD 8.49 billion in 2024 to USD 44.09 billion by 2030, at a CAGR of 32%**. This growth is increasingly driven by generative AI and emerging agentic AI systems that enable automation, decision-making, and intelligent workflows. As adoption accelerates, demand for skilled AI professionals continues to rise globally.

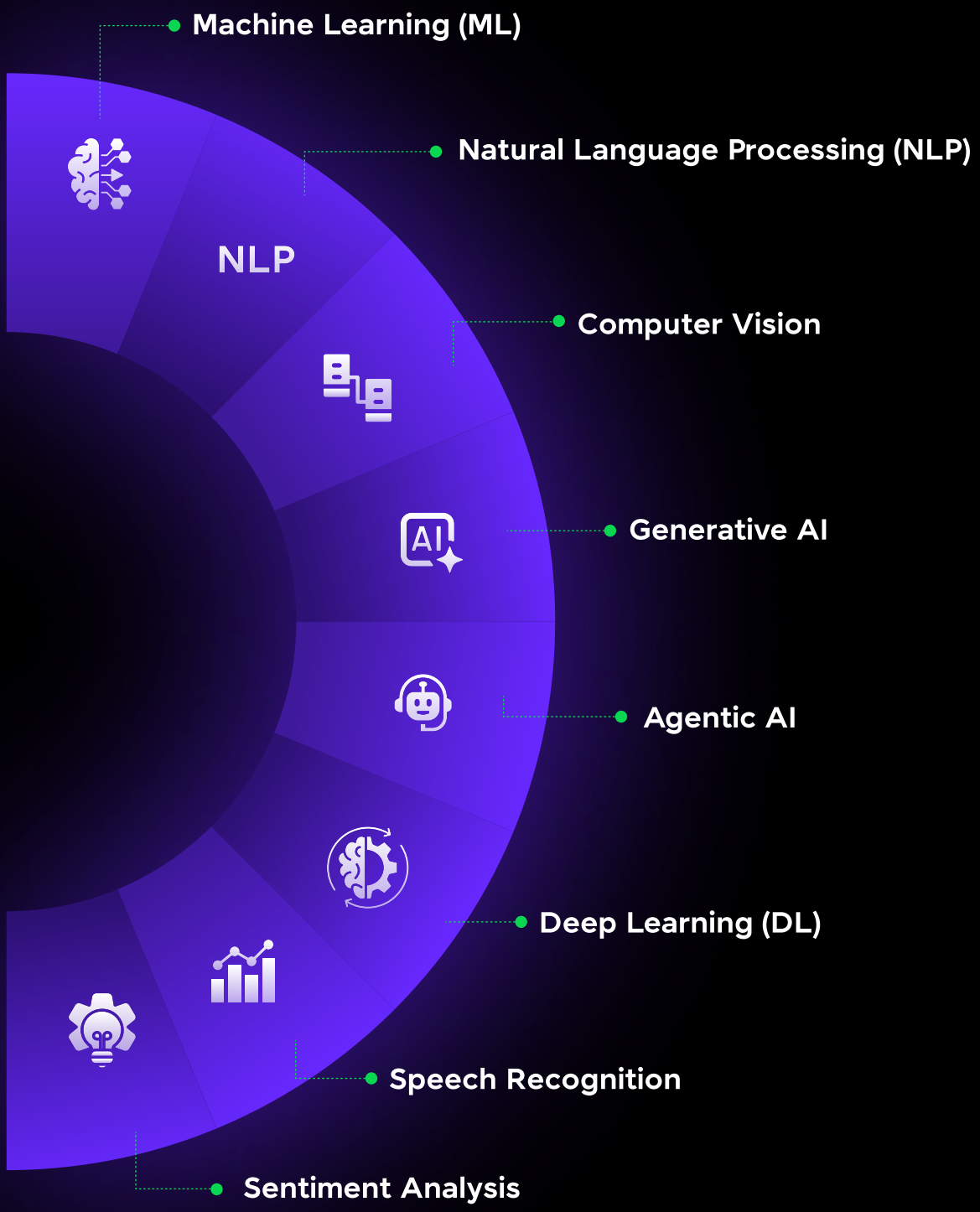
Artificial Intelligence (AI) & Machine Learning (ML) - Industry 5.0 Market, 2018-2030



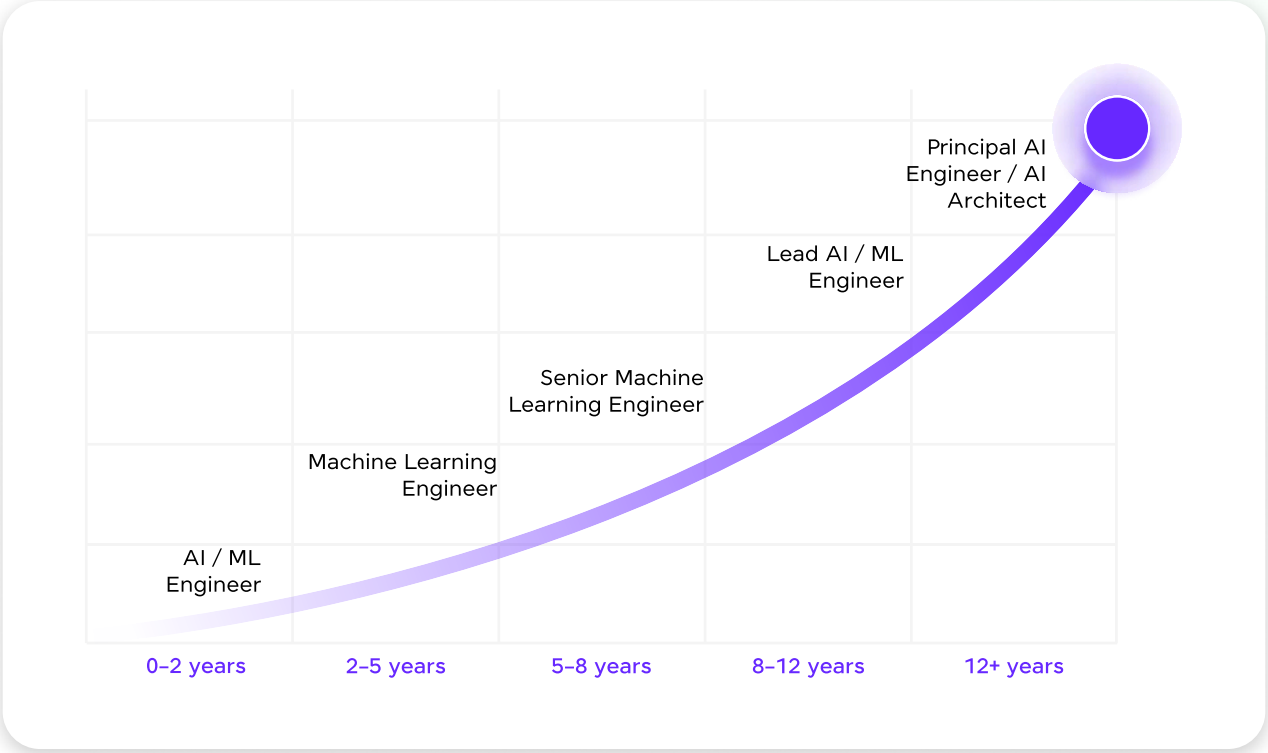
From Traditional AI to Agentic AI

Traditional AI	Agentic AI
Focuses on making predictions	Does more than predict, it plans and acts
Works in fixed, step-by-step flows	Adapts as the situation changes
Usually built for one specific task	Can handle multi-step tasks on its own
Has limited understanding of context	Understands context and remembers interactions
Needs human guidance to move forward	Works more independently with minimal input

Explore a Range of Career Domains in AI ML



Here’s What Your Career Growth Can Look Like



Course Overview

The programme offers a comprehensive immersion into modern AI/ML and advanced agentic AI systems, enabling learners to understand how intelligent systems are built, trained, deployed and scaled for real-world applications. It builds strong competence across the core pillars shaping today's AI ecosystem, including:

Data & ML Foundations

data handling, feature engineering and machine learning models

Deep Learning

neural networks, training techniques and optimization

NLP & Generative AI

transformers, LLMs and prompt engineering

Agentic AI Systems

RAG pipelines, AI agents and multi-agent workflows

Deployment & AI Engineering

APIs, MLOps and production-ready systems

What Makes This Programme Stand Out

- Industry-aligned AI/ML curriculum for real-world applications
- Globally recognized certification
- Expert-led sessions by AI practitioners and industry professionals
- Project-focused training with real-world use cases and system design
- Career support with placement assistance and interview preparation
- Access to 1000+ corporate hiring partners
- Live, interactive learning with lifetime recorded access
- Hands-on experience with LLMs, RAG systems, and AI agents
- Exposure to production tools, MLOps, and deployment workflows
- Easy EMI options available

Who This Programme is For

- Freshers looking to build a career in AI/ML and data-driven roles
- Career switchers entering AI, machine learning or data science domains
- Developers and engineers upskilling in AI, deep learning and LLMs
- Professionals exploring generative AI, RAG systems
- Anyone serious about building real-world AI projects and becoming job-ready

Program Curriculum

Module 1: AI Landscape & Current Trends

• Understand the evolution of AI from machine learning to generative and agentic AI, including key industry applications and system types
• Explore AI career paths and domains such as ML, deep learning, NLP, and agentic systems, and how to choose the right track
• Learn AI project strategy, portfolio building, and gain exposure to tools like Google AI Studio and Claude

Module 2: Python Programming

• Build strong programming fundamentals including data types, control structures, functions, and working with strings, lists, and dictionaries.
• Master advanced concepts like OOP, file handling, modules, APIs, and working with JSON and regular expressions.
• Work with Python tools such as Jupyter Notebook and VS Code for hands-on coding and development.

Module 3: SQL & Data Warehousing

• Learn database fundamentals including normalization, keys, transactions, and relational database concepts
• Write efficient SQL queries using joins, subqueries, aggregations, window functions, and query optimization techniques
• Understand data warehousing concepts and integrate SQL with Python using modern database systems

Module 4: Pandas for Data Handling

• Work with DataFrames for data loading, cleaning, filtering, and transformation across real-world datasets
• Perform operations like merging datasets, GroupBy analysis, pivot tables, and date-time handling
• Apply data manipulation techniques through mini projects using Pandas

Module 5: Data Visualization using Python

• Create visualizations using bar charts, line plots, histograms, scatter plots, and advanced charts like heatmaps and pairplots
• Understand data storytelling, time series visualization, and choosing the right chart for different data scenarios
• Build styled visual outputs using Matplotlib and Seaborn for analysis and presentation

Module 6: Data Cleaning & Transformation

• Perform exploratory data analysis including handling missing values, outliers, and understanding data distributions
• Apply feature engineering, encoding, scaling, and normalization techniques for model readiness
• Handle imbalanced datasets and implement feature selection methods using Scikit-learn

Module 7: Classical Machine Learning

• Understand supervised and unsupervised learning techniques including regression, classification, clustering, and dimensionality reduction
• Work with algorithms like Logistic Regression, KNN, Decision Trees, Random Forest, XGBoost, KMeans, and PCA
• Evaluate and optimize models using metrics, cross-validation, and pipeline-based workflows for deployment readiness

Module 8: Neural Networks Foundations

• Learn the fundamentals of neural networks including perceptron, activation functions, and loss functions
• Understand forward and backward propagation along with optimization concepts
• Build a strong foundation for deep learning using NumPy

Module 9: Deep Learning Core

• Understand key concepts like bias-variance tradeoff, overfitting, underfitting, and model training strategies
• Apply regularization techniques such as dropout, L2, batch normalization, and early stopping
• Work with deep learning frameworks like PyTorch and TensorFlow

Module 10: Deep Learning with PyTorch

• Learn PyTorch fundamentals including tensors, autograd, and computation graphs
• Build and train neural networks using datasets, dataloaders, and training loops
• Understand GPU-based training and model validation techniques

Module 11: Computer Vision (CNN)

• Understand computer vision fundamentals including image processing, CNN architectures, and convolution operations
• Work with image preprocessing, augmentation, object detection concepts like IoU and MAP
• Build vision-based models using tools like OpenCV, Roboflow, and YOLO

Module 12: NLP with Neural Networks

• Learn text preprocessing techniques along with Bag of Words, TF-IDF, and word embeddings
• Understand sequence modeling and architectures like RNN, LSTM, and GRU
• Build NLP applications for tasks like text classification using tools like NLTK and SpaCy

Module 13: Transformer Models

• Understand limitations of traditional models and the need for transformer architectures
• Learn attention mechanisms, encoder-decoder models, and architectures like BERT and GPT
• Fine-tune transformer models and evaluate performance using metrics like BLEU and ROUGE

Module 14: Introduction to LLMs

• Understand large language models, their architecture, and real-world applications across industries
• Explore popular LLMs and how they are used in modern AI systems
• Gain hands-on exposure using tools like OpenAI and Gemini

Module 15: Prompt Engineering

• Learn prompt design techniques including zero-shot, few-shot, and chain-of-thought prompting
• Work with structured outputs, function calling, and guardrails for reliable responses
• Build real-world applications like chatbots using tools such as OpenAI API, LangChain, and LlamaIndex

Module 16: RAG Systems

• Understand RAG architecture including retrievers, generators, embeddings, and vector databases
• Work with chunking strategies, metadata filtering, and techniques to reduce hallucinations
• Build and evaluate RAG systems using tools like FAISS, Pinecone, Weaviate, and LangChain

Module 17: API Development

• Understand API fundamentals including REST architecture, request-response handling, and endpoint design
• Build APIs using Flask and FastAPI, and integrate machine learning models into production endpoints
• Test, document, and manage APIs using tools like Postman

Module 18: MLOps & Deployment

• Learn experiment tracking, model versioning, and CI/CD workflows for machine learning systems
• Understand model monitoring, drift detection, and deployment pipelines using Docker and cloud platforms
• Work with tools like MLflow, DVC, AWS for production-ready AI deployment

Module 19: AI System Design

• Design scalable AI systems including data pipelines, model pipelines, and inference workflows
• Understand trade-offs between batch vs real-time systems, latency vs accuracy, and cost optimization
• Learn to architect reliable, efficient, and production-grade AI systems

Module 20: AI Agents

• Understand agent architectures including planning, execution, tool usage, and memory systems
• Build agent-based workflows with API integrations and multi-step reasoning
• Develop real-world agentic applications using frameworks like LangChain Agents

Module 21: Multimodal AI

• Learn how AI systems process and combine multiple data types like text and images
• Understand vision-language models, image captioning, and multimodal assistants
• Build applications using tools like CLIP, LLaVA, and Gemini

Module 22: Responsible AI

• Understand AI ethics, bias, fairness, and governance in real-world systems
• Learn about data privacy, security risks, and responsible AI practices
• Analyze case studies to understand ethical challenges and solutions

Build With the Tools That Power Modern AI

AIML Stack (Core + GenAI Foundations)

Programming & Data

Python

NumPy

Pandas

LLM Platforms

OpenAI

Google Gemini

Hugging Face

Visualization & Databases

Matplotlib

Seaborn

MySQL / PostgreSQL

LLM Frameworks

LlamaIndex

Machine Learning & Deep Learning

Scikit-learn

PyTorch

TensorFlow

RAG & Vector Databases

FAISS

Pinecone

Weaviate

NLP & Computer Vision

NLTK

SpaCy

OpenCV

Roboflow

Agent Frameworks

LangChain
Agents

GenAI Foundations

OpenAI API /Gemini

Tooling & Automation

Function
Calling - MCP

Multimodal AI

CLIP

LLaVA

Deployment & LLMOps

Docker

AWS

MLOps

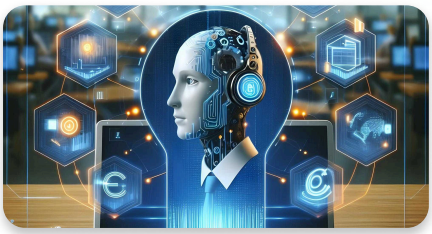
Flask

FastAPI

MLflow

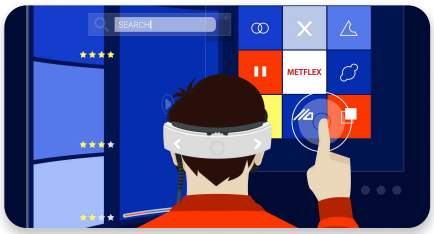
Build Industry-Ready Projects That Get You Hired

Build an End-to-End Predictive Analytics System with ML and NLP



Develop a Smart Recommendation System using ML and Deep Learning

Create an AI-Powered Document Intelligence System with NLP and RAG

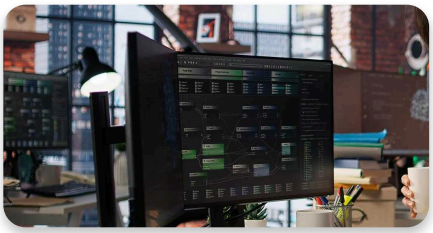


Build a Computer Vision System for Real-World Detection and Insights

Develop a Conversational AI Assistant with Context Awareness

Build a Multimodal Sentiment Analysis System using Text and Vision

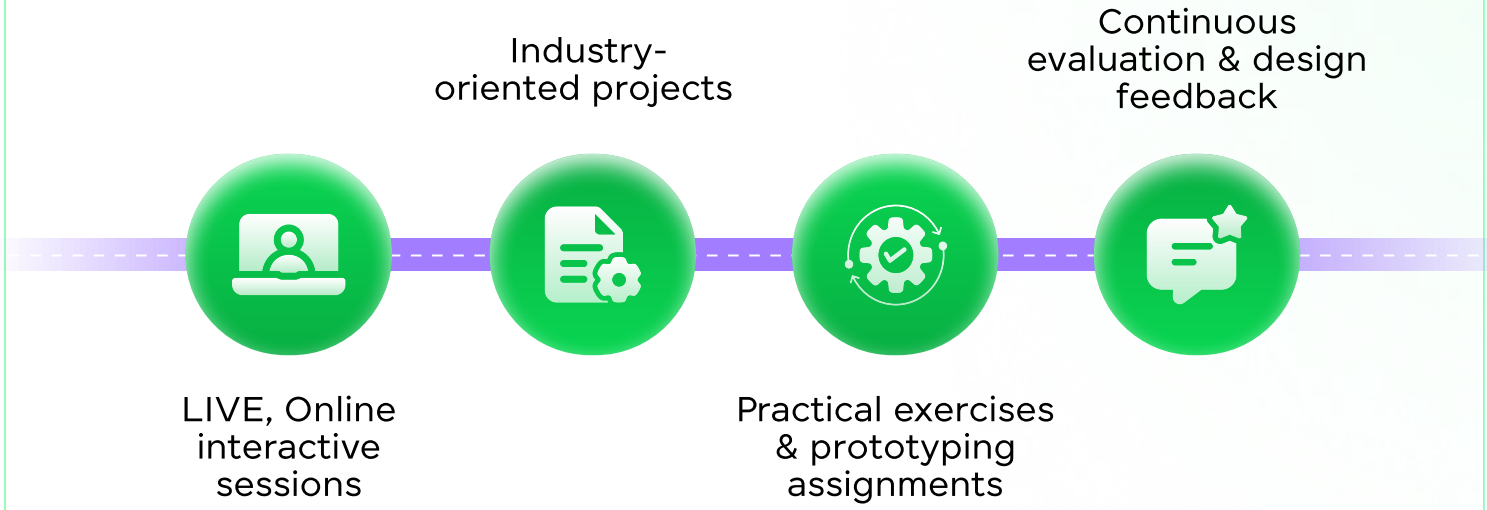
Create a Time Series Forecasting and Anomaly Detection System



Build an Enterprise Multimodal Knowledge Assistant using RAG

Develop a Multi-Agent Research and Report Generation System

Programme Delivery Framework



Key Details About The Programme

Duration: 6 Months

Schedule : Weekend/ Weekday

Delivery: Live Online Sessions + Recorded Sessions

Eligibility:

- Working professionals transitioning into AI, ML, or data-driven roles
- Graduates seeking their first job in AI/ML or data science
- Students & freshers aspiring to become AI/ML engineers

Benefits of an HCL GUVI Certification

Industry-Recognized Certification
Validates your AI/ML skills for real-world applications.

Hiring Advantage
Strengthens your profile and improves job opportunities across AI and data roles.

Career Growth Opportunities
Opens pathways across AI, machine learning, data science, and emerging AI domains.

Get Personalized Guidance from Top Industry Mentors

Sukanya Chandrasekaran
NLP & AI Architect
Tiger Analytics

Mohammed Afsar Meera
AI/ML Computational Science Specialist
accenture

Vinish Vivek
Agentic AI Engineer
EDAG

Harshit Panchal
Founder & CEO

Shreya Shahi
Software Development Specialist
NTT DATA

From Where They Started to Where They Are Now

Sai Sudharsan S G

Fresher

AI/ML engineer

ATNA TECHNOLOGIES

Aashifa

Fresher

AI developer

Praveen R

Non-Tech

AI ML engineer

JET

Our Learners' Success Highlights

Naveen mohan

"From the beginning, HCL GUVI provided a clear roadmap for the AI & ML course and taught everything from the basics of Python and SQL to Machine Learning, Deep Learning, and NLP. The mentors were friendly and supportive, and they guided us well during projects by sharing ideas and giving the right direction whenever needed."

Gopinath C

"The AI & ML course at HCL GUVI is very useful for my future. The team explained the concepts clearly, cleared my doubts, and supported me throughout the learning process. The CodeKata platform was especially helpful, with coding questions ranging from beginner to advanced levels, which improved my problem-solving skills."

Achuraj D

"I completed the Artificial Intelligence and Machine Learning program at HCL GUVI. Aswin, our mentor, handled the sessions very well and explained concepts clearly. The support from HCL GUVI during the course was excellent, with quick responses to queries and emails. I'm completely satisfied with the learning experience and happy that I chose the right path for my course."

Fee Structure

Particulars	Amount
Programme Fee	**₹94,999 (Inclusive of taxes)**

- EMI options available
- Refund policy as per T&C
- Referral rewards applicable

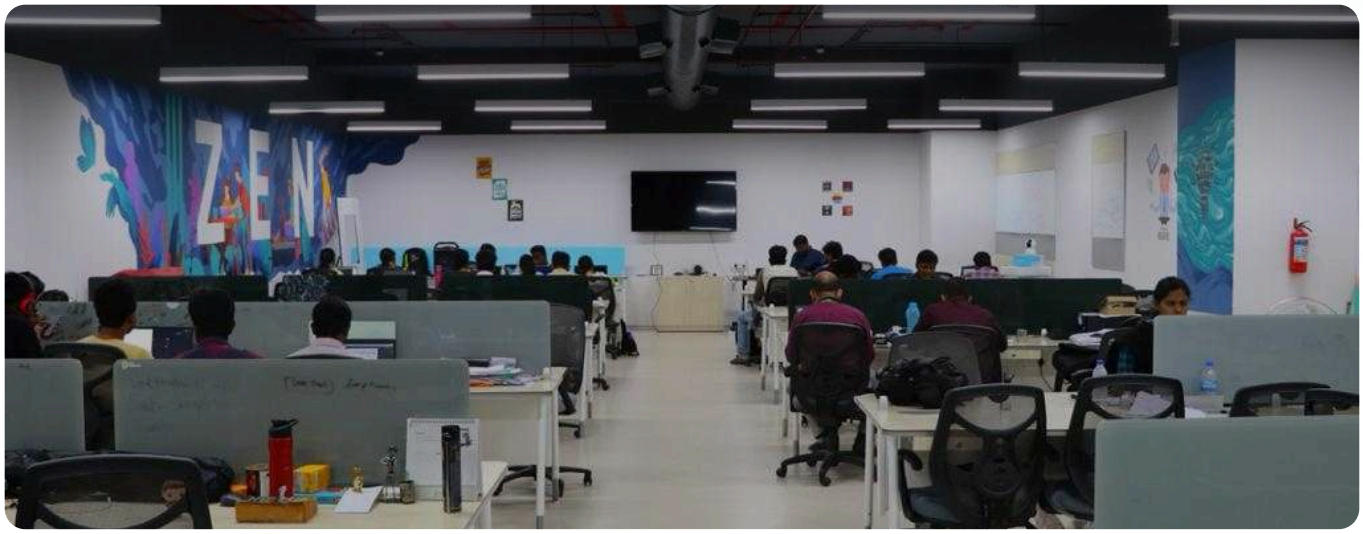
Become a job-ready AI ML Engineer with hands-on projects and industry-standard tools. Learn machine learning, deep learning, and GenAI workflows at just **₹7,974* /Month**

Train intelligent models. Build autonomous systems. Scale with AI.

Start Your Journey With HCL GUVI

About HCL GUVI

HCL GUVI is the technology upskilling arm of the HCL Group, focused on preparing learners for emerging digital careers. Incubated by IIT-Madras and IIM-Ahmedabad, HCL GUVI delivers industry-aligned programmes in 20+ vernacular languages and has enabled 4.5M+ learners across India. With 1000+ hiring partners, a vernacular-first model and placement support, HCL GUVI connects education with employability in the modern tech ecosystem.



We are Accredited by

AICTE All India Council for Technical Education

N·S·D·C National Skill Development Corporation
Transforming the skill landscape

NASSCOM®

ISO 27001-9001 CERTIFIED

Google for Education Partner

Built by Passionate Educators

Arun Prakash

CEO & Founder at GUVI

20+ years of Technical Expertise

PayPal Honeywell

& more

Built 7 Products from Scratch Mentored 1000+ students Hosted 200+ sessions & 25+ webinars

Bala Murugan

Co - Founder at GUVI

17+ years of experience with IT industry

PayPal

Technologist with 9+ years of Entrepreneurial experience & Member of the Syllabus Sub-Committee at Anna University

By the Numbers: HCL GUVI's Impact

4 M+
Learners Upskilled

2000+
Industry Mentors Onboard

24+
Languages Supported

300+
Colleges hosted Developer Bootcamps

1000+
Corporate hiring partners across India

5000+
Students Placed in First Jobs

Recognitions from Industry & Government

Digital India Contributor
Recognized by Times Now

Titan LEAP Partner
3 years of collaboration

AI for India 2.0
National skilling initiative

500K+ Women Trained
Through GUVI x SAWiT.AI

Tamil Nadu Hackathon
5500+ students empowered